

Deliverable 4 - Plotting

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1. Plot 1 - Original data

File: plots/plot_1_original_quality_distribution.png

This chart shows the distribution of the original quality scores before transforming the target. It is useful because it shows which scores are common and which scores are rare. We can observe that middle scores are the most frequent, while extreme scores are less frequent.

2. Plot 2 - Feature-engineered data

File: plots/plot_2_engineered_quality_group_distribution.png

This chart shows the distribution of quality_group, the final target used for classification. It is useful for understanding the imbalance between low quality, medium quality, and high quality. We can observe that the classes are not perfectly balanced, which supports the use of macro F1-score.

3. Plot 3 - Model performance

File: plots/plot_3_model_performance_macro_f1.png

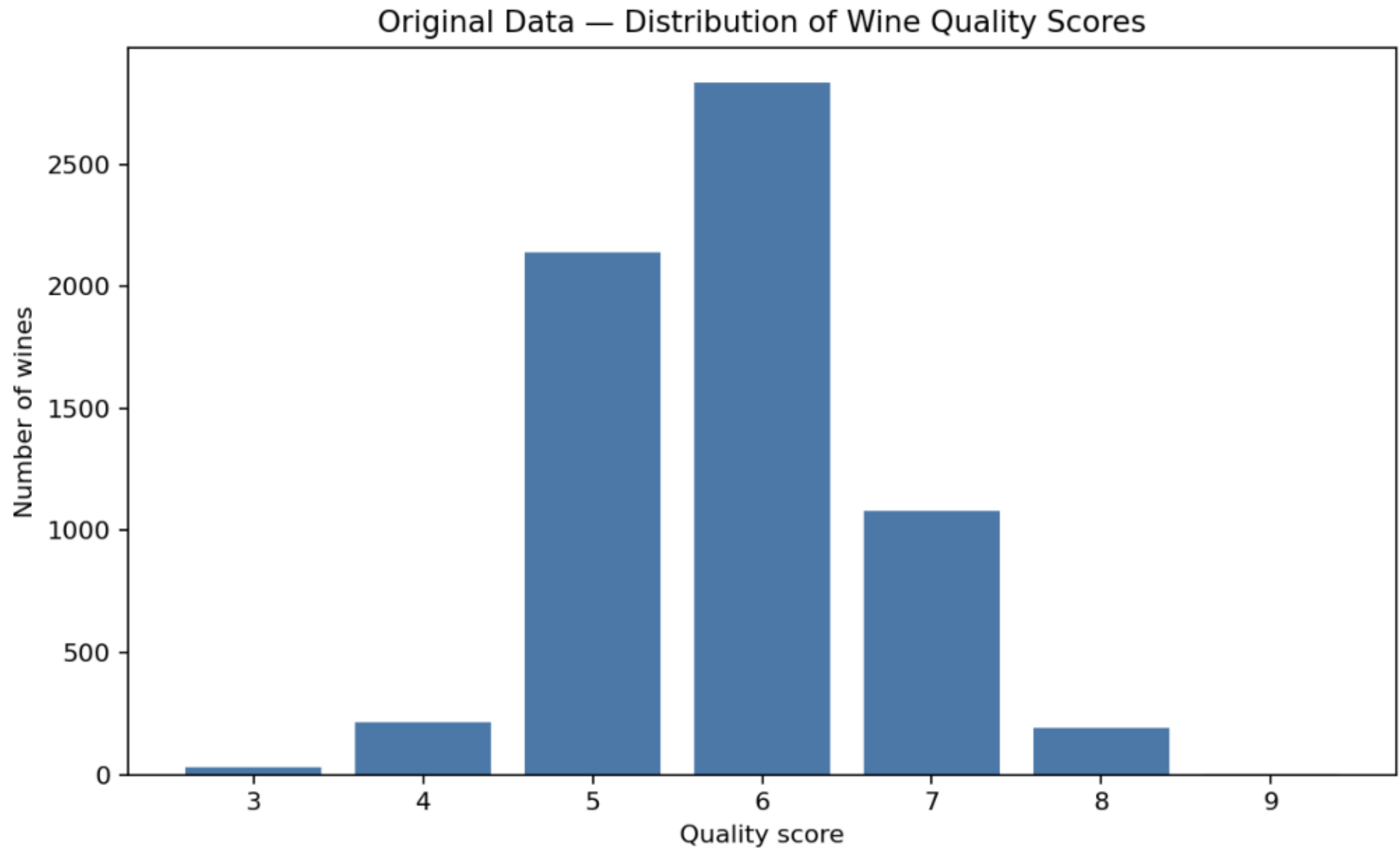
This chart compares the three models using the main metric, macro F1-score. The values come directly from results/model_metrics.csv.

Results used:

model	accuracy	precision_macro	recall_macro	f1_macro
Logistic Regression	0.582	0.585	0.539	0.550
Random Forest	0.732	0.748	0.712	0.726
SVM	0.627	0.637	0.576	0.589

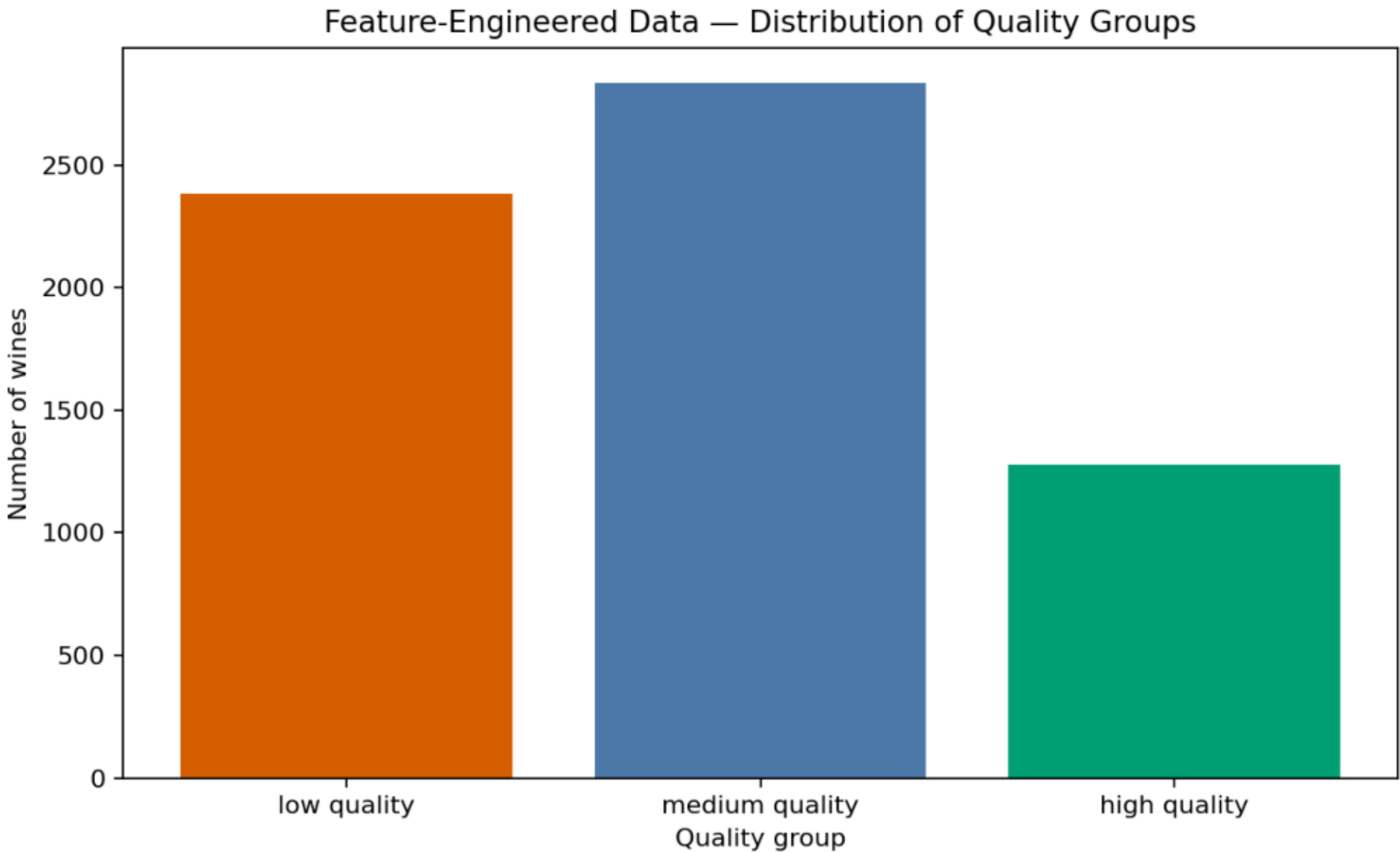
This chart is useful for a simple model comparison. It shows that **Random Forest** is the best model in this first comparison, but this is still a POC result and not a final conclusion.

Plot 1 - Original quality distribution



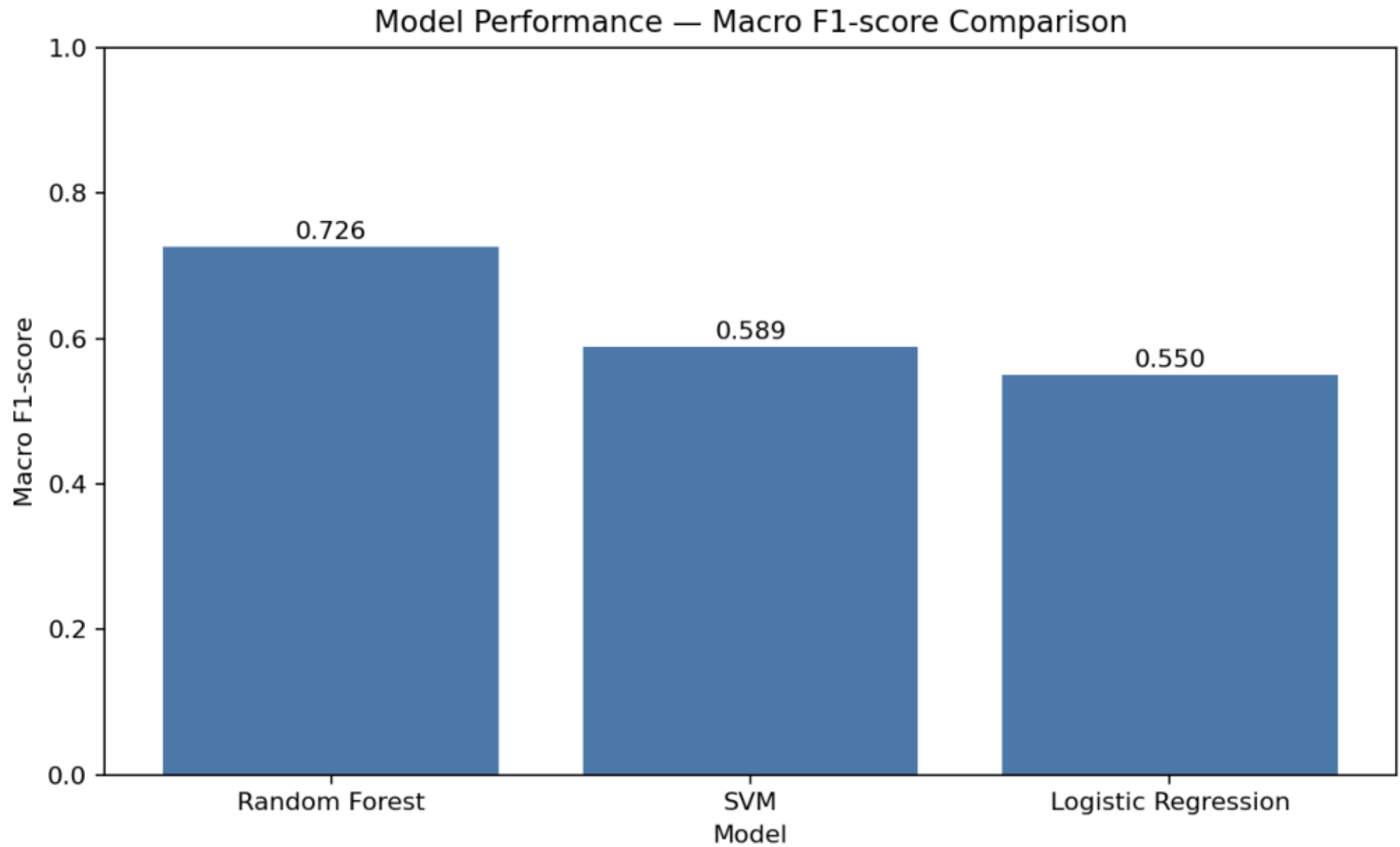
Distribution of original wine quality scores.

Plot 2 - Quality group distribution



Distribution of the engineered target quality_group.

Plot 3 - Macro F1 model comparison



Macro F1-score comparison using real model metrics.